

The Socratic Method: Cross-Resolution Supervision for Papyrus Surface Segmentation

ANONYMOUS AUTHOR(S)
SUBMISSION ID: TBD

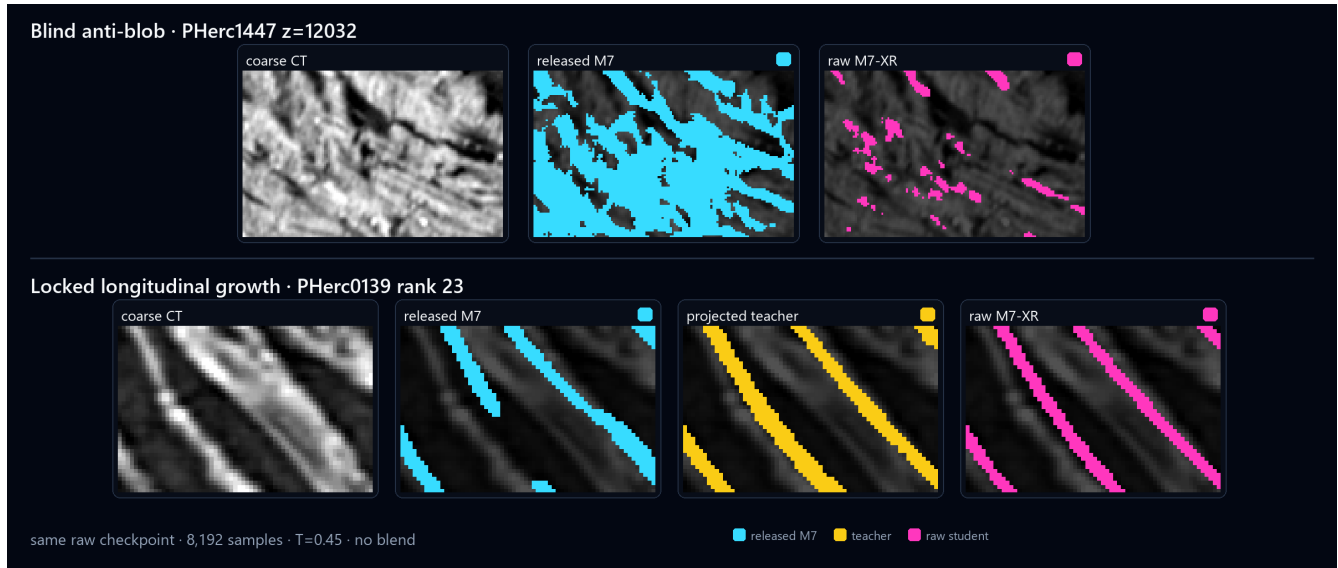


Fig. 1. One raw checkpoint answers two opposing failure modes. Both rows begin with the identical CT crop and released M7 state-of-the-art baseline. The top row shows blind PHerc1447 de-blob behavior; the bottom row adds the projected fine teacher and shows longitudinal growth on locked PHerc0139. Rendered panels in this draft show the earlier v31 student at $t = 0.45$; Sec. 5.6 gives the panel provenance and the outstanding regeneration against the shipped model.

Coarse-resolution papyrus surface segmentation fails in two opposing ways: faint sheets disappear, while nearby wraps merge into blobs. We present a teacher-student method for Herculaneum papyrus data in which an approximately $2.399 \mu\text{m}$ native-fine model interrogates the released $9.362 \mu\text{m}$ M7 segmenter. The resulting student remains an ordinary, single M7 residual-encoder nnU-Net at inference. Fine-teacher predictions enter the coarse frame as soft occupancy with an explicit validity field, and the student is held to M7 by one masked function-space term that acts where the teacher is silent or confidently agrees. Trained over twelve scrolls for a full 250,000-exposure schedule, the student reaches 0.6808 macro-scroll Dice on a frozen 689-row benchmark against released M7's 0.5541 under matched inference, and 0.7130 with eight-way mirror test-time augmentation at a preregistered operating threshold. We also report what did not survive. An earlier form of this method carried five further training terms—medial crest recall, a teacher-background separation shell, one-sided M7 preservation, and fixed-route then dynamic widest-path connectivity—together with a global parameter trust region. Given a corpus of adequate breadth and no trust projection, every one of those terms is measurably neutral or harmful. The trust region itself was roughly 82 times smaller than the displacement the successful recipe requires, and its saturation is what concealed that fact for as long as it did. Those negative findings, and the measurement failures behind them, are reported in the appendices rather than omitted.

CCS Concepts: • **Computing methodologies** → **Neural networks**.

Additional Key Words and Phrases: volumetric segmentation, knowledge distillation, papyrus, topology-aware learning, curve completion, ablation

ACM Reference Format:

Anonymous Author(s). 2026. The Socratic Method: Cross-Resolution Supervision for Papyrus Surface Segmentation. *ACM Trans. Graph.* XX, X, Article XX (September 2026), 16 pages. <https://doi.org/10.1145/nnnnnnnn.nnnnnnnn>

1 Introduction

“I neither know nor think that I know.”
—Socrates, in Plato's *Apology* 21d [Plato nd]
SOCRATES: I am wiser than this man; he fancies he knows something, although he knows nothing—
DARRYL, SOCRATES' FRIEND: *fuck him up socrates*
—leon (@leyawn) [leon (@leyawn) 2015]

The objects behind this work are the Herculaneum papyri, a library of scrolls carbonized by the AD 79 eruption of Mount Vesuvius. Carbonization preserved the rolls but left them too brittle to open and the writing trapped inside tightly wound coils. Synchrotron micro-CT can record their internal structure without touching the object; recovering a page then requires locating the papyrus sheet in three dimensions and *virtually unwrapping* it to a plane [Seales et al. 2016; Vesuvius Challenge 2026b]. A scroll is one continuous sheet, metres long, visiting many adjacent turns. Consequently, two pieces separated by only a few CT voxels may be distant along the page, while a short gap in the volume may interrupt one continuous line of text.

Segmenting the papyrus surface is the first geometric commitment in a scroll-unwrapping pipeline. A missed surface interrupts a sheet that may otherwise be traceable over a long distance. An overgrown prediction is at least as dangerous: it can weld nearby turns into one component and make a locally plausible surface globally wrong. A useful segmentation method must therefore recover faint structure and remove false connections at the same time. An overlap score alone does not adequately describe this tension.

The Vesuvius data provides a second, complementary source of information. A model operating on finer CT can resolve local evidence that is ambiguous after downsampling, while the deployed segmentation and geometry stack still consumes the coarser grid. We treat this mismatch as a cross-resolution dialogue. The fine model proposes soft occupancy; the released M7 model contributes a function that is already effective; an explicit loss term exposes where those views are inconsistent. The student must reconcile them.

This is the origin of our title. In the Socratic account, a speaker who professes little knowledge reveals inconsistencies in an expert's position and thereby points toward a better one. Our fine teacher is similarly not deployed as a superior replacement. It asks local questions of the expert coarse model. The answer is a revised M7 that stands alone at inference.

A method arrived at by subtraction. This paper describes a smaller method than the one we first built, and that difference is a substantial part of its content. An earlier version of this work supervised the student with seven weighted terms: soft occupancy and Dice, medial crest recall, a teacher-background separation shell, a masked M7 function anchor, one-sided M7 positive preservation, and a dynamic widest-path connectivity term, all under a global parameter trust region. Each term was a plausible local answer to an observed failure, and the stack as a whole did improve on released M7. When the corpus was later widened from one scroll to twelve, the trust projection removed, and the declared schedule actually run to completion, single-term ablation showed that five of the six auxiliary terms were neutral or harmful, and that the surviving one—the M7 function anchor—accounted for the measurable gain. We describe the reduced method in the body and the discarded terms in Appendix A. Appendix B treats the trust region separately, because it is not simply another retired term: while it was active it made the others unmeasurable, and it is the reason the original stack survived scrutiny for as long as it did.

We present four contributions:

- a hash-pinned $2\mu\text{m}$ -to- $9\mu\text{m}$ teacher-supervision recipe that can be executed independently of our geometry repository;
- an objective reduced by single-term ablation to soft occupancy and one masked M7 function anchor, reported together with the five discarded terms and the measurements that discarded them;
- an evaluation protocol that runs the declared schedule to completion, quantifies patch-sampling uncertainty by paired cluster bootstrap, and measures seed noise with a full-horizon replicate before crediting any difference; and

- a separately measured, additive 2D curve fitter that reconnects only short gaps able to answer geometric and cross-plane objections (Appendix C).

The 250,000-exposure schedule and its independent gate review are complete. We report every retained milestone, the selected checkpoint, the inference-time levers that were preregistered before scoring, and the negative results that reduced the objective to its present form.

2 Related Work

Segmentation. U-Net architectures and their self-configuring nnU-Net variants are widely used for dense volumetric segmentation [Isensee et al. 2021; Ronneberger et al. 2015]. Our deployed network follows the released M7 residual-encoder nnU-Net from the Vesuvius Challenge project exactly; we alter its supervision and optimization rather than adding an inference-time branch. This preserves compatibility with the existing coarse-volume pipeline.

Teacher supervision. Knowledge distillation transfers information from a teacher into a student by matching soft outputs or intermediate functions [Hinton et al. 2015]. Most such methods assume teacher and student predictions share a sampling grid. Our teacher observes a finer CT volume, so its occupancy and validity must be pulled back into the student's physical frame. The resulting targets represent partial volume rather than only a hard class. Our surviving constraint on the incumbent model is also a distillation term, but it is masked rather than global: it acts where the teacher is silent or confidently agrees and is switched off in the band where teacher and incumbent disagree, which is precisely the region the method exists to revise.

Topology-aware objectives. Skeleton and centerline losses preserve thin connected structures that can be under-represented by region overlap metrics [Shit et al. 2021]. The medial axis transform instead describes a shape by centers and the radii of their maximally inscribed disks or spheres; LSMAT casts this construction as a least-squares problem to make the representation less brittle under noisy boundaries [Rebain et al. 2019]. This center-radius factorization is attractive for papyrus, because motion along the medial set extends a sheet whereas increasing radius thickens it toward a neighboring wrap. We built three successive objectives on that distinction—crest recall, a separation shell keyed to the crest, and a widest-path connectivity term over teacher-medial corridors—and none of them survived measurement on a corpus of adequate breadth. Appendix A reports those results in full, including the respect in which the underlying geometric argument still appears correct and the respect in which it turned out to describe a property of our training corpus rather than of the objective.

Curve completion. Classical thinning extracts one-pixel skeletons from binary images [Zhang and Suen 1984]. Curve completion can then be posed as endpoint pairing under geometric continuity. For wound papyrus, however, a locally smooth completion can cross from one wrap to the next. Our postprocess therefore combines 2D tangents with an umbilicus-aware radial test, neighboring-plane corridors, explicit anti-merger and crossing gates, and connection persistence across slices (Appendix C).

Vesuvius models and data. Our experiments build on the public Vesuvius Challenge CT data, the released M7 surface model, and a Villa native-fine teacher [Vesuvius Challenge 2026a,b; Villa Model Authors 2026]. The final paper will replace these artifact-level entries with the authoritative model and dataset citations associated with the public release.

3 Overview

Fig. 2 summarizes our pipeline. We first run a native-fine teacher in its approximately $2.399\ \mu\text{m}$ CT frame. Its probabilities and support are projected into the $9.362\ \mu\text{m}$ grid as soft training fields (Sec. 4). We initialize every candidate from the exact released M7 checkpoint and train all parameters, unconstrained in parameter space, for a declared 250,000-exposure schedule. The output is one ordinary M7 network. At inference we use only its raw class-1 probability; there is no teacher and no M7 probability blend. The shipped configuration adds eight-way mirror test-time augmentation at a preregistered operating threshold, which changes inference cost but not the weights.

After selecting a threshold, an optional C postprocess operates on each axial slice (Appendix C). It proposes Bezier completions between skeleton endpoints, but paints a join only when local curve geometry, radial displacement, neighboring slices, and connection-track support agree. Segmentation training and curve completion are measured separately so an improvement in one stage cannot hide a failure in the other.

Two engineering details prevent accidental leakage or model substitution. First, held-out morphology is used only for evaluation; no gate corpus contributes a training target. Second, the M7 initializer is accepted only if both its byte size and SHA-256 match the released artifact. A new candidate may resume its own interrupted optimizer state, but may not use an earlier student as learned initialization.

4 Cross-Resolution Supervision

4.1 Student Architecture and Data

Our student is a six-stage 3D residual-encoder nnU-Net with feature counts (32, 64, 128, 256, 320, 320), one CT input channel, and two output classes. It has 102,349,770 trainable parameters. Every parameter is retrained, but initialization is accepted only when the released M7 checkpoint matches its recorded SHA-256; the check is fail-closed and includes the original segmentation heads.

The training corpus contains 16,200 registered 192^3 rows: 15,432 training and 768 validation, drawn from twelve training scrolls and two held-out validation scrolls. Of the training rows, 10,072 carry anti-aliased soft teacher targets and 6,128 are human-labelled rows passed through as hard targets. We draw 250,000 sample exposures from this schedule; “250,000” counts repeated draws, not unique rows.

Corpus breadth is the single largest lever we measured, and it is worth stating plainly because it displaced most of our loss engineering. An earlier configuration of this method trained on 4,096 rows from one scroll. Widening to twelve scrolls, with no other change to the objective, moved the frozen benchmark further than any auxiliary loss term we designed (Appendix A).

Our held-out validation manifest contains registered PHerc0814 and PHerc1451 patches, frozen before this study. We evaluate every 10,000 exposures and retain a model-only snapshot at each of the 25 milestones. Because the milestone response is a shallow random walk rather than a monotone curve (Sec. 5), we neither assume the final checkpoint is best nor select on the best single milestone without accounting for seed noise.

4.2 Fine-Teacher Occupancy

The fine teacher provides one target in the student’s physical frame: soft occupancy $q \in [0, 1]$, obtained by an anti-aliased pullback from the $2.399\ \mu\text{m}$ grid, with an independent validity mask V recording where the teacher is defined at all. Occupancy estimates how much of a $9.362\ \mu\text{m}$ voxel is filled by papyrus, and retaining it as a fraction rather than a hard vote preserves the partial-volume evidence at a sheet boundary that motivated the cross-resolution setup.

The occupancy ceiling, and why it is a corpus property. A thin but real continuation commonly occupies only 25–45% of a coarse voxel. Soft cross-entropy is correctly calibrated when it drives such a voxel toward $p = 0.25$ – 0.45 , and the result may then sit below the segmentation threshold. Read as a property of the loss, this observation says that occupancy supervision has a ceiling and needs a separate sheet-existence target; that reading produced the medial crest term described in Appendix A. Read as a property of the data, it says something narrower: on a single-scroll schedule, the fraction of continuation-band voxels the student can push over a usable threshold is small. The second reading is the correct one. On the present corpus the same diagnostic does not fire—a clear majority of continuation-band validation voxels already sit above the lowest gate-passing threshold—and the crest term, measured there, does not help. We retain occupancy alone.

4.3 Objective

Let p be the student’s surface probability, q the projected fine-teacher occupancy, V its validity field, and p_0 the released M7 probability from a frozen copy of the initializer carried through training. Our loss is

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(p, q) + \mathcal{L}_{\text{Dice}}(p, q) + 0.5 \mathcal{L}_{\text{KL}}(p, p_0). \quad (1)$$

All terms are evaluated only where their validity fields permit. Every other term we have implemented carries weight zero, and there is no parameter-space projection; Appendices A and B record what those terms were and how they were measured.

Soft occupancy. Cross-entropy and Dice both take the fractional target q rather than a thresholded label, and both are reduced per sample before averaging over the samples that contain any supervised voxel. The per-sample reduction is deliberate: a dense human-labelled patch and a sparse native-teacher patch otherwise contribute in proportion to their valid voxel counts, which lets one supervision source dominate a mixed corpus.

M7 function anchor. The teacher is locally more informative than M7, not globally complete, and across most of any training volume it says nothing at all. We therefore anchor the student to the released function wherever the teacher is silent or confidently agrees, and

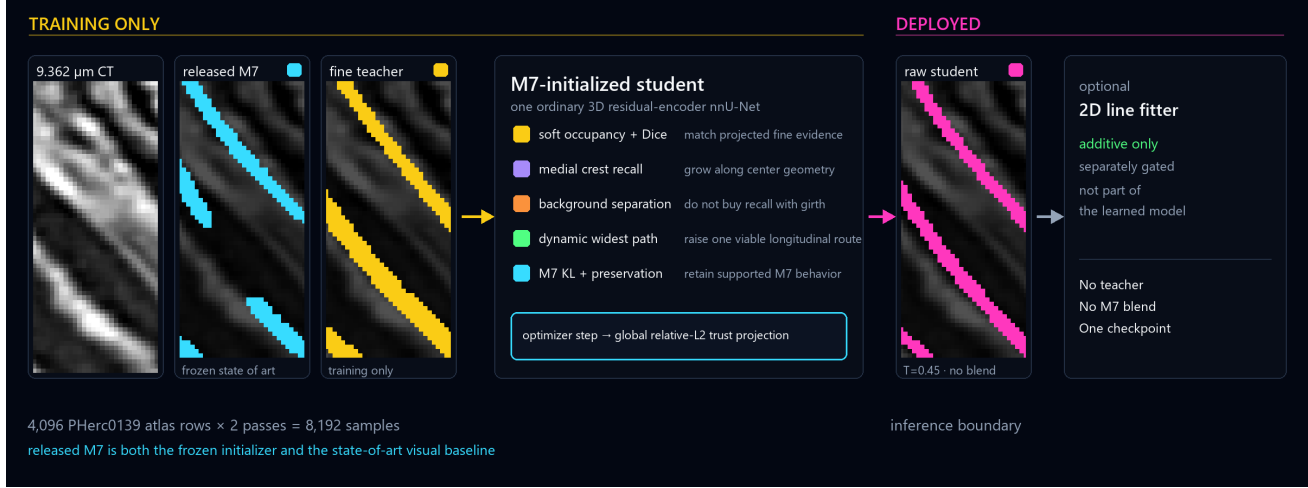


Fig. 2. Method overview with a registered crop. The released M7 appears both as the frozen initializer and as the state-of-the-art visual baseline. Training-only components are separated from the deployed raw student, and the optional 2D fitter is outside the learned model and evaluated separately. The rendered schematic dates from the earlier seven-term configuration, so the medial and separation losses and the post-update trust projection it depicts are no longer part of the method; they are documented in Appendices A and B.

leave it free everywhere else. Let $H = \{q \geq 0.5\} \cap V$ be the teacher's hard positives, $h_x = \mathbf{1}[x \in H]$, $m_x = \mathbf{1}[p_{0,x} \geq 0.5]$, and let D_2 denote radius-two dilation. The anchor mask is

$$K = (\bar{V} \setminus D_2(H)) \cup \{x \in V : h_x = m_x \wedge (q_x \leq 0.1 \vee h_x = 1)\}, \quad (2)$$

and the term is

$$\mathcal{L}_{KL} = \frac{1}{|K|} \sum_{x \in K} D_{KL}(\text{Ber}(p_{0,x}) \parallel \text{Ber}(p_x)). \quad (3)$$

Three properties of K carry the term's behaviour, and all three were needed before it measured positive. First, unknown space is anchored, so the student cannot invent structure where nothing has been observed. Second, a radius-two corridor around known teacher positives is carved back out of that unknown region, so M7 is not permitted to freeze the immediate neighbourhood of a sheet the teacher wants grown. Third, inside known space the anchor acts only where M7 and the teacher agree *and* the teacher is confident, meaning $q \leq 0.1$ or $q \geq 0.5$. Without this last restriction, a partial-volume voxel at $q = 0.3$ with M7 below its own decision boundary counts as agreement on background, and the anchor then pulls down exactly the continuation band the occupancy target is trying to raise. The direction $D_{KL}(\text{Ber}(p_0) \parallel \text{Ber}(p))$ is mode-covering with respect to M7: it penalizes the student for abandoning M7's mass, not for adding its own where M7 is unopposed.

Deep supervision. Both occupancy terms and the anchor are applied at the standard nnU-Net decoder outputs with normalized weights proportional to $1, 1/2, 1/4, \dots$ and zero weight on the coarsest output. Soft occupancy is resized trilinearly and its validity is true only where every contributing fine cell is valid.

4.4 Optimization

We use SGD with learning rate 10^{-3} , Nesterov momentum 0.99, weight decay 3×10^{-5} , gradient clipping at 12, and a polynomial decay of exponent 0.9 declared over the full 250,000-exposure horizon. The batch size is three with no gradient accumulation, under bfloat16 autocast. Seed 1203 fixes the sample order and augmentation sequence.

Declaring the schedule at its true length matters for comparability. The learning rate at exposure n depends on the declared horizon, so a run stopped early is a different optimizer trajectory from a run that was declared short, and milestone scores taken while the rate is still near its initial value describe a hot random walk rather than a converged model. Every arm in this study, including the ablations of Appendix A, declares the same 250,000 horizon so that matched milestones are comparable even when an arm is terminated early.

There is no trust region and no post-update projection. Earlier versions of this method projected the parameter vector into a relative- L^2 ball around released M7 after every update. That constraint, its saturation, and the measurement it invalidated are the subject of Appendix B.

5 Results

5.1 Evaluation Protocol

Our primary measurement is macro-scroll Dice on a frozen 689-row benchmark drawn from the two held-out scrolls: the unweighted mean of the per-scroll pooled Dice, so a scroll contributes equally regardless of how many rows it supplies. The benchmark manifest, its row set, and its threshold grid were fixed before this study and are hash-checked at evaluation time.

Three protocol choices matter for interpreting what follows. First, the operating threshold is declared rather than fitted per candidate: $t = 0.35$ is the primary endpoint and $t = 0.30$ the single preregistered

secondary point, chosen a priori because it is the calibrated argmax at 21 of 25 milestones. A per-candidate calibrated threshold is reported as a diagnostic only, since selecting it inside the same rows inflates the score. Second, uncertainty is measured by paired cluster bootstrap: sampling units are support-anchor chunks within a scroll rather than individual rows, because many validation rows share an anchor, and a candidate and its reference are resampled with identical weights so their difference is paired. Such an interval describes patch sampling and says nothing about seed noise, which we measure separately. Third, a promoted candidate must additionally pass a blind six-cube morphology gate on PHerc1447 and a frontier veto: its reference-relative bridge pixels may not exceed 1.20 times the reference curve interpolated at matched skeleton coverage, with no extrapolation outside the measured coverage overlap.

5.2 Milestone Behaviour and Seed Noise

All 25 milestones completed their frozen audits and were gate-eligible at $t = 0.35$. The response is not a curve that rises and plateaus. Over the eleven late milestones at or beyond 150,000 exposures, the mean fixed-threshold macro Dice is 0.6613 with a standard deviation of 0.0107, and adjacent milestones swing by several times that deviation in both directions. Batch three with momentum 0.99 random-walks in a flat basin.

The preregistered late-selection rule chooses 200,000 exposures at 0.680787. For context rather than selection, the global maximum over all milestones is 0.6816 and the exact endpoint is 0.6693, so a rule that simply took the last checkpoint would have lost roughly a standard deviation of milestone noise.

We ran a second full-horizon replicate under an identical recipe with a different seed in order to know how much of that spread is not attributable to the method at all. Its root-mean-square difference from the primary run across the 25 matched milestones is 0.0295, falling to 0.0191 over the late band. This number is the reason we do not claim duration superiority: the gap between our selected milestone and an early milestone of the same recipe is smaller than the difference two seeds of that recipe produce. It is also the yardstick every ablation in Appendix A is measured against.

5.3 Comparison with Released M7

Table 1 reports the frozen benchmark under matched inference. Against released M7 evaluated identically and at the same threshold, the student gains 0.1266 macro Dice, a 22.9% relative improvement. The comparison is not an ensemble: inference runs the raw student alone, with no teacher forward pass and no probability blend.

5.4 Inference-Time Levers

Two levers that change no weights were preregistered before scoring, together with the rule that a claim above 0.70 requires a positive paired lower bound after Holm correction and non-negative deltas on both scrolls. Eight-way mirror TTA at $t = 0.30$ reaches 0.7130 [0.7012, 0.7248], a paired gain of +0.0230 [+0.0195, +0.0265] over the same weights without TTA, positive on both scrolls. At the primary endpoint $t = 0.35$ the same weights reach 0.7038.

Table 1. Frozen 689-row benchmark, macro-scroll Dice. TTA is eight-way mirror test-time augmentation on unchanged weights. The student's t values are preregistered; M7's calibrated $t = 0.20$ is fitted in-sample and therefore favours the baseline.

Model	Inference	Macro Dice
Released M7	$t = 0.35$	0.5541
Released M7	TTA, $t = 0.30$	0.5875
Released M7	TTA, calibrated $t = 0.20$	0.5926
Ours (200k)	$t = 0.35$	0.6808
Ours (200k)	TTA, $t = 0.35$	0.7038
Ours (200k)	TTA, $t = 0.30$	0.7130

We also evaluated uniform weight averages over late milestones. They reduce variance without shifting the level: no narrow late average beat the selected milestone, and only a wide average spanning 100,000 to 250,000 exposures edged past it, by less than the seed band. We report those variants and do not promote them.

At the shipped operating point the blind six-cube gates and the frontier veto pass: foreground ratio 0.799 against the reference, with 4,945 reference-relative bridge pixels against a 7,983 ceiling at the measured skeleton coverage of 0.782. A Kaggle-style composite on the same rows rises from 0.6401 to 0.6521, with the topology component rising from 0.3880 to 0.4016 and thereby meeting the 0.3776 advisory that the no-TTA configuration missed.

5.5 Held-Out Human Labels

On PHerc0500P2, the one held-out set with human rather than model-derived labels, every variant we measured scores between 0.476 and 0.497 composite: the spread is smaller than the differences we are trying to resolve. Released M7 with TTA at its calibrated threshold remains ahead there at 0.5015. These are absolute scores only and not a gain claim, because released M7 saw those labels during its own training; we report them because a flat result on the only human-labelled holdout is a real limit on what the frozen benchmark establishes.

5.6 Figure Provenance

Every rendered panel in this draft—the teaser, Fig. 3, Fig. 4, and both galleries in Sec. 8—shows the earlier v31 student at $t = 0.45$, not the model reported in Table 1. Those panels were produced for the earlier configuration and have not yet been regenerated. They remain informative about the qualitative failure modes the method addresses, and the registered comparison grammar is unchanged, but they should not be read as depicting the shipped weights. Regenerating the panel set against the selected checkpoint is outstanding release work and is listed in Sec. 6. The quantitative gate review of the shipped model, including its six-cube panels, is complete and lives in the release record.

5.7 Interpretation

The result we are most confident in is a level: under matched inference the student is far enough above released M7 that no threshold, seed, or milestone choice within our measured spread reverses the

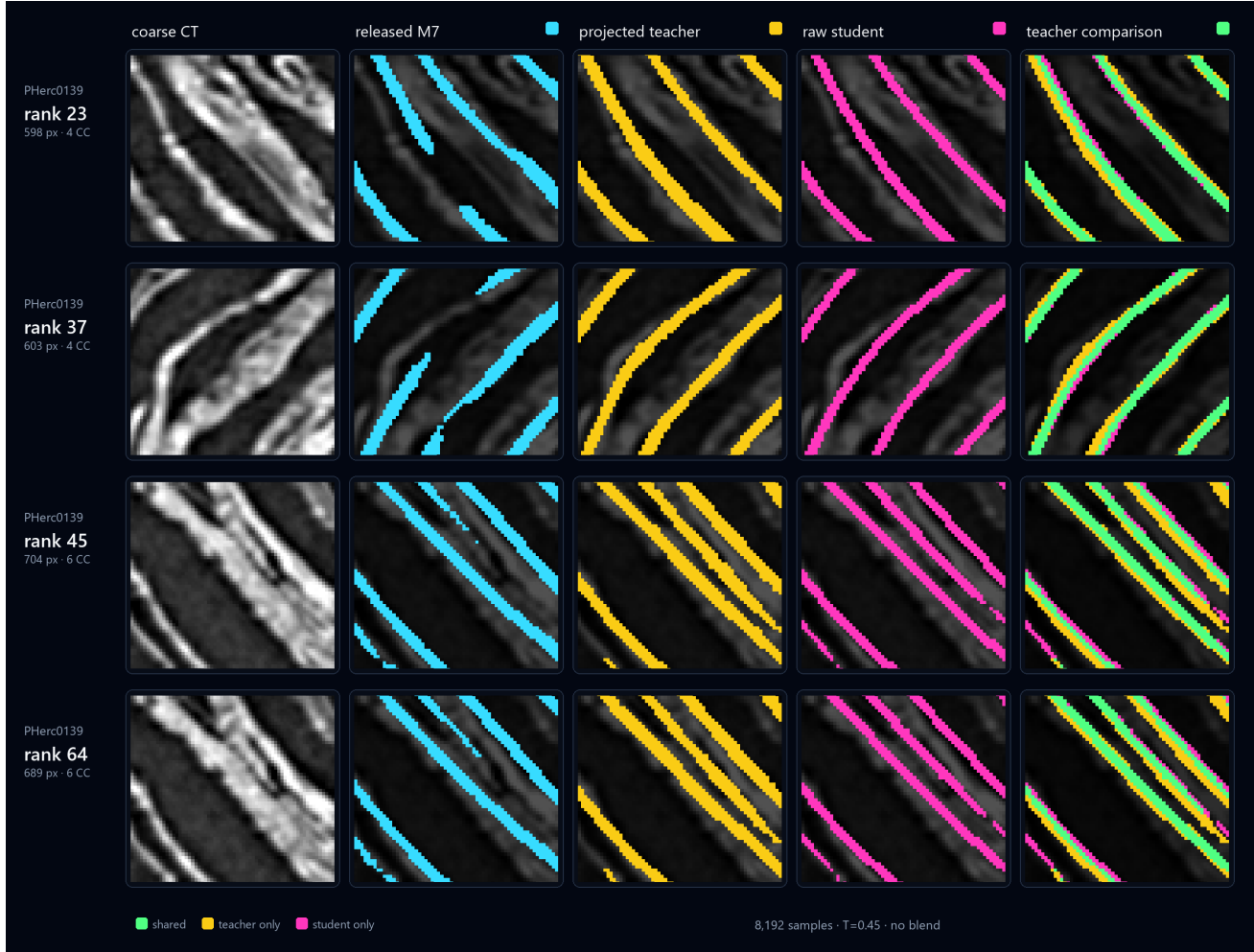


Fig. 3. Locked PHerc0139 comparisons at a fixed operating point. Columns show the same CT crop, released M7, projected fine-teacher target, raw student, and a two-sided teacher comparison. Green denotes shared foreground, yellow teacher-only foreground, and magenta student-only foreground. Student panels show the v31 configuration; see Sec. 5.6.

ordering. The results we hold loosely are the ones inside that spread. Duration beyond the late band, the choice among late milestones, and the weight-average variants all sit within seed noise and are reported as unresolved rather than settled. Both gate corpora remain model-derived rather than ground truth, so the morphology review constrains the selected checkpoint without certifying it, and the flat human-labelled holdout keeps that caveat in view.

6 Limitations

Fine-teacher predictions are model-derived soft supervision rather than human ground truth, and both gate corpora are likewise model-derived. Our guards therefore constrain a checkpoint without certifying it. The one held-out set with human labels is flat across every variant we measured, and released M7 remains ahead there, so the frozen benchmark establishes an ordering under its own supervision rather than a general one.

Milestone-to-milestone variation is comparable to the differences that selection rules are asked to resolve. We measured seed noise with one full-horizon replicate and report anything inside that band as unresolved, but a single replicate is a weak estimate of a variance, and we have not repeated the shipped configuration a third time.

The ablations in Appendix A are negative results measured under one corpus, one optimizer, and one horizon. They establish that those terms do not earn their place in this configuration. They do not establish that no version of a medial, separation, or connectivity objective could help under a different corpus or schedule, and the medial tail floor in particular remains untested at scale because the corpus it needs does not exist in the required form.

Test-time augmentation improves the shipped number at eight times the inference cost. We report both the augmented and unaugmented figures because that trade is a deployment decision, not a property of the model.

Rendered figures in this draft show the earlier v31 student and have not been regenerated against the shipped weights (Sec. 5.6); doing so is outstanding release work. The downstream postprocessors of Appendix C keep provenance pins at the earlier operating threshold and must be re-qualified before they are combined with the shipped inference recipe.

The curve fitter has a different failure mode from the model. A completion may be locally smooth and persistent across slices but still join the wrong wrap near the core. Radial, third-component, and cross-sheet gates reduce this risk but cannot prove global correctness. The postprocess must remain independently measured and its additions must remain recoverable during qualification.

Finally, exact replay requires large, separately licensed artifacts. The training code, recipes, hashes, and export scaffolding are public-repository material, but the validation corpus, teacher atlases, M7 weights, and student checkpoints require dedicated artifact hosting and resolved upstream terms.

7 Conclusion

We presented a cross-resolution supervision method for improving the released M7 papyrus-surface segmenter without changing its deployed architecture. A fine teacher exposes local inconsistencies as soft occupancy; a masked function anchor holds the student to the incumbent wherever the teacher is silent or confidently agrees, and releases it everywhere else. Under matched inference the resulting student improves on released M7 by 0.1266 macro-scroll Dice on a frozen benchmark, and by more with an inference-time augmentation that leaves the weights untouched.

The larger lesson of this work is subtractive. We began with seven weighted training terms, each a defensible answer to an observed failure, and finished with three. What removed the others was not a better idea but a better measurement: a corpus wide enough for a shape prior to be worth less than data, a schedule run to its declared length, and the removal of a parameter-space constraint that had been quietly making every ablation uninformative. The terms we discarded were not obviously wrong when we added them, and they were not shown to be wrong by argument. They were shown to be wrong by measuring them in a regime where they could have been right.

The same design principle survives downstream: the 2D fitter may propose a connection, but radial, merger, cross-sheet, crossing, and persistence gates force it to answer a sequence of geometric objections before adding pixels. A useful surface model must learn to grow longitudinal structure without buying that recall through thickness or cross-wrap mergers, and the honest way to know whether a given mechanism does that is to give it a fair chance to fail.

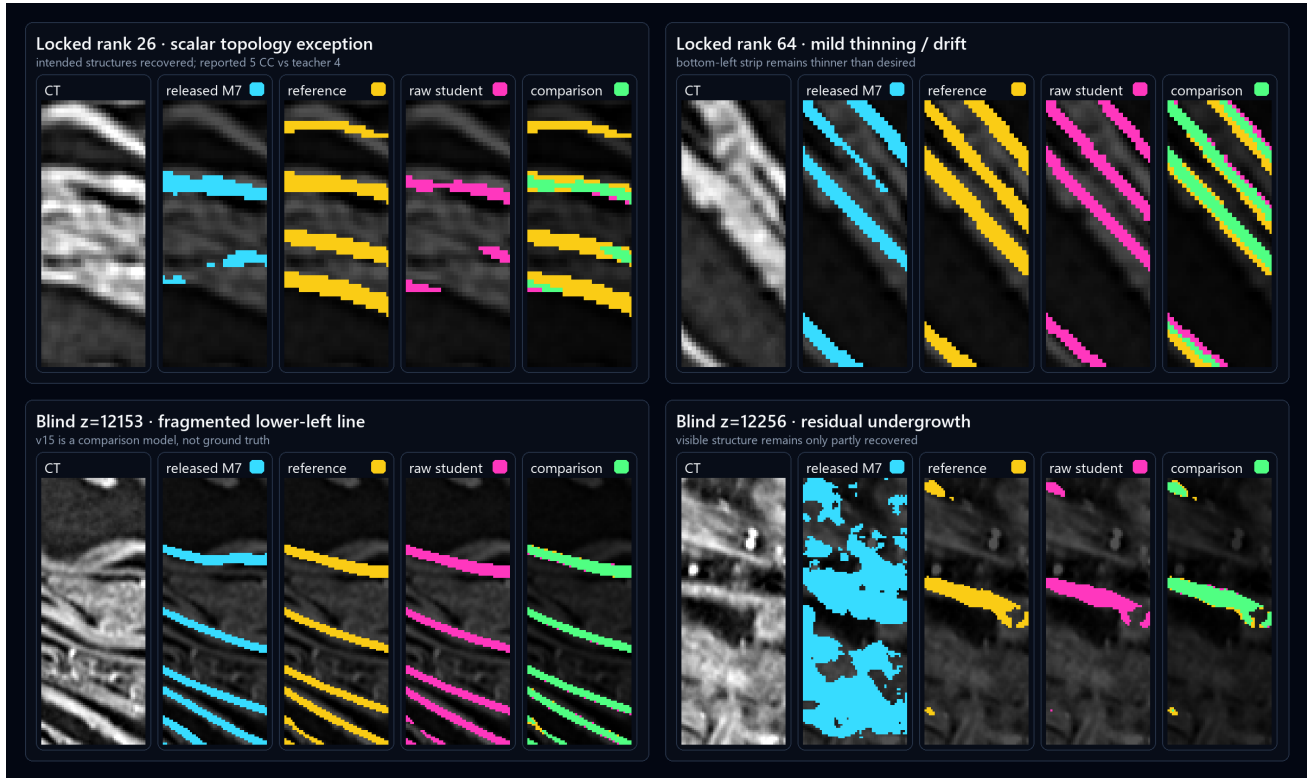


Fig. 4. Known edge cases: a component-count proxy that disagrees with the visible structure, mild drift or thinning on locked sheets, and blind PHerc1447 views that remain undergrown or fragmented. Every row includes CT, released M7, the available model-derived reference, the raw student, and a two-sided comparison. Yellow reference masks are not ground truth. Student panels show the v31 configuration; see Sec. 5.6.

8 Qualitative Gallery

The following pages collect the clearest qualitative results we have rendered. They are presentation views, not additional samples or a hidden selection stage: every crop appears in the committed locked-16 or fixed PHerc1447 review corpus.

Both galleries show the earlier v31 student at $t = 0.45$ rather than the checkpoint reported in Table 1, and have not been regenerated against the shipped weights; see Sec. 5.6. They are included because the failure modes they illustrate—longitudinal growth on locked slices, and de-blob behaviour on blind cubes—are the ones the method targets, and because the released M7 baseline shown beside every panel is unchanged.



Fig. 5. Locked PHerc0139 growth gallery. Every registered crop is shown as CT, released M7, projected fine-teacher target, and the raw student. These examples emphasize learned longitudinal regularity while keeping the released state of the art visible.



Fig. 6. Blind PHerc1447 anti-blob gallery. The released M7 foreground is cyan and the raw student's foreground is magenta, each on the identical CT slice. No fine teacher exists for these views.

Table 2. The earlier seven-term objective. “Weight” is the shipped value in that configuration. Only the first two terms and the M7 function anchor remain. Section references point to the measurement that retired each term.

Term	Weight	Status	See
Soft cross-entropy	1.0	retained	Sec. 4
Soft Dice	0.25	retained, reweighted	Sec. 4
M7 function anchor	0.5	retained	Sec. 4
Medial crest recall	1.0	retired	A.4
Separation shell	2.0	retired	A.3
M7 preservation	1.0	retired	A.5
Dynamic connectivity	0.03125	retired	A.6
Trust region	—	retired	App. B

A What We Also Tried

The objective in Eq. 1 has three terms. An earlier version of this method had seven, listed in Table 2. This appendix records what those terms were, why each seemed justified, and what happened when they were measured in a regime where they could have been right. We report them because the reduction was the most informative result of the project, and because a reader deciding whether to build a similar objective deserves the negative half of our evidence.

A.1 Why the Stack Survived as Long as It Did

The terms were not retained because they had been shown to work. They were retained because, in the regime where they were developed, nothing could be shown either way, and a term that shows no measurable effect is easy to keep. Three properties of that regime were jointly responsible.

First, the parameter-space trust region was saturated: the post-update projection was active on essentially every optimizer step, so no term could move the function far enough to demonstrate either benefit or harm. Appendix B treats this in detail because it is the load-bearing failure.

Second, the promotion gate rewarded the behaviour the terms produced. Its dominant criterion required that at least 95% of the M7 voxels already judged teacher-correct survive within a one-voxel neighbourhood of the student’s prediction. That is a retention test, and preservation and the anchor are retention terms. A thinner, better student fails it. Cross-referencing the artifacts of that era, no shipped term scored above the no-term baseline on this gate; the only intervention that moved it was an inference-time blend with M7, which was itself later withdrawn as confounded. A companion anti-blob check compounded the problem by pinning its reference to an older prediction grid with a foreground cap, which bounded any candidate at roughly 39% of the foreground of a previously accepted model and made growth scoreable only as a violation.

Third, the training corpus had narrowed to 4,096 rows from a single scroll. A shape prior is most attractive exactly when data is scarce, and least testable for the same reason.

A fourth, procedural point belongs here. The terms were added incrementally and evaluated as a stack against the previous stack, so what was tuned was the interaction, not the terms. Leave-one-out arms were specified for five of them and never trained. Preservation

Table 3. Single-term ablations on the wide corpus with no trust region, best frozen milestone, macro-scroll Dice. The seed replicate differs from the baseline only in its random seed and bounds what any single arm can claim.

Arm	Macro Dice
Occupancy only (CE + Dice)	0.6542
Occupancy only, seed replicate	0.6455
+ separation shell 0.5	0.6349
+ M7 function anchor 0.5	0.6769

is the clearest casualty of this ordering: it was always paired with the separation shell that offsets its cost, so the question of what it does alone was never asked until it was asked directly (Sec. A.5).

A.2 Single-Term Ablations

Once the corpus was widened and the trust region removed, each candidate term could be added individually to a common baseline. Table 3 reports arms trained from released M7 under the optimizer of Sec. 4, all declaring the same horizon and all terminated at the same milestone, differing only in the named flag.

The anchor is the one term that clears its own noise floor, and it is the one term in Eq. 1. The separation arm lands below the baseline. The two occupancy-only arms differ from each other by more than most effects we were trying to detect, which is why every later comparison is expressed against a measured seed band rather than against zero.

A.3 Separation Shell

The shell placed background cross-entropy on a two-voxel dilation of the teacher-supported positives, intersected with confidently empty teacher space. The intent was a local fence: attack unsupported girth and short cross-wrap welds while leaving the occupancy term free to grow faint material elsewhere.

Measured as a dial, it does what it claims. On a separate release-ladder sweep, raising its weight from 2 to 3 cut reference-relative bridge pixels by roughly half. It also cost about sixteen points of reference skeleton recall, against a tolerance of one, and lowering it toward zero inverted both effects. What the sweep does not show is any movement of the underlying trade. Compared at matched coverage rather than at a fixed threshold, a model trained with the shell sits on the same bridge-versus-coverage curve as one trained without it; the shell selects a point on that curve, and so does the segmentation threshold, which is free. We retired it because a training-time term that duplicates a deployment-time dial should be the dial.

A.4 Medial Crest Recall

The crest term supervised sheet *existence* separately from amount of material. The fine teacher’s medial surface was extracted with the center-radius view of LSMAT [Rebain et al. 2019], projected to the coarse grid with a max operator so that a thin center survives downsampling, and rewarded through a per-sample recall term evaluated only on crest voxels. Slice-wise radii were deliberate: pairing 2D centers with a 3D distance transform would substitute spheres for disks and collapse the medial surface toward a curve.

Its premise was the occupancy ceiling of Sec. 4. That premise is corpus-dependent, and on the wide corpus the diagnostic that was supposed to trigger the term does not fire: the fraction of continuation-band validation voxels already above the lowest gate-passing threshold is comfortably above the level at which a ceiling would be indicated. The term was accordingly never triggered for a wide-corpus arm of its own. Where it was measured, on the narrower release ladder, raising its weight reduced bridge burden by roughly a fifth but cost five points of skeleton recall against the same one-point tolerance, and the rendered panels showed it refilling broad mass rather than extending thin lines. Global recall pressure turned out to be the wrong shape for the problem: it buys length by buying width.

A focused variant survives as untested rather than refuted. Instead of rewarding all crest voxels, it applies a one-sided hinge to the weakest fifth of them, so gradient vanishes once a selected voxel clears a floor. It had the best burden-versus-recall behaviour of the medial family in its one trial, and it cannot be evaluated on the present corpus at all: the term requires a medial sidecar on every training row, and the human-labelled rows of the wide corpus can never carry one. Testing it means building a crest-complete corpus first, which changes the corpus and therefore needs its own control arm.

A.5 M7 Positive Preservation

Preservation applied a one-sided foreground cross-entropy to voxels where the frozen M7 predicted foreground inside a two-voxel corridor around teacher positives. Its stated job was to stop de-blobbing from deleting M7-supported sheet, and it had the best surviving motivation of any retired term: our students beat M7 on overlap and on merge behaviour but lose to it on topology on the one human-labelled holdout, which is the deficit preservation is shaped to fix.

It was also, of the seven, the only term never measured in isolation at scale, so we measured it. Three arms were preregistered against the shipped recipe, each declaring the full horizon and terminating at a common milestone: the original hard form at its shipped weight; a one-sided soft-floor form that resists only where the student has fallen below M7; and, conditional on a rim-thickening trigger, the hard form paired with a reduced separation shell. Kill criteria, a seed-derived tie margin, and the thickening trigger were fixed before any arm ran.

All three were rejected, and the failure mode was the one the preregistration named. Pooled paired deltas against the shipped recipe over five matched milestones were -0.0480 for the hard form, -0.1117 for the soft floor, and -0.0228 for the shell-paired arm, every interval excluding zero. Each arm failed the blind six-cube morphology gate at two consecutive milestones on interior fraction and maximum thickness. Foreground ran between 1.3 and 2.3 times the reference where the shipped recipe runs at 0.71, and reference-relative bridge pixels ran an order of magnitude above it. The frontier veto could not even be evaluated: at the coverage these models reach, they lie beyond the end of the reference curve, which is itself the finding. None of the arms was inert—the term

contributed between 13% and 22% of the committed training loss—so these are measured negatives rather than absences.

The shell-paired arm is the informative one. It halves the damage on every axis and still fails every gate. Read together, the three arms say that preservation's cost is paid in girth and that the separation shell of the original stack existed to pay it. Reconstructing that pair means reintroducing a term we retired in Sec. A.3 for duplicating a free deployment-time dial.

A.6 Dynamic Connectivity

The connectivity term asked, existentially, for some viable route through an audited teacher-medial corridor joining two or more disconnected M7 components. Its precursor prescribed one minimum-off-axis route and raised the weakest tenth of its logits; that version tied the incumbent at the smallest weight and degraded ordinary validation monotonically at larger ones, which is what motivated replacing a prescribed path with a bottleneck on whichever path the student currently supports. The mechanism is sound and we still regard it as the most interesting of the retired terms.

It is nonetheless inapplicable to the corpus that produced our result, for a structural reason rather than a measured one. An event requires a full-volume M7 prediction to define the component-contact pins, and that prediction exists for exactly one of the training scrolls. The frozen atlas contains 149 events, all from that scroll; on the wide corpus roughly three percent of rows carry any event at all, so the term is close to inert by construction and its weight cannot be interpreted. Producing the missing full-volume predictions is days of inference, and until that exists the honest status is untested at breadth, not refuted.

A.7 Terms Proposed and Rejected Later

Two further candidates were measured after the reduction and are recorded for completeness. A hardened Dice term, taking the thresholded label the benchmark actually scores in addition to the soft target, lost to the shipped recipe at both matched milestones with intervals excluding zero, and carries a documented confound: human rows are already binary, so their Dice weight doubles. Dropping the human-labelled rows entirely lost at the earlier milestone and won narrowly at the later one, inside the seed band, and was not promoted.

A.8 The Retired Configuration, for Reference

Table 4 preserves the completed duration ladder of the seven-term configuration on its single-scroll corpus, as originally reported. Its best calibrated macro-scroll Dice was 0.58293 at 7,168 samples, a gain of roughly 0.02 over released M7. The present method gains 0.1266 on the frozen benchmark. The entire seven-term stack was worth about a sixth of what corpus breadth and a completed schedule were worth, which is the ratio we would most like a reader to take away.



Fig. 7. Supervision fields from the retired seven-term configuration, shown on actual training data at dynamic event 138. Projected soft occupancy is retained by the present method; the sparse medial crest, the radius-two separation shell, the teacher-medial corridor, the component-contact pins and the free anchors supplied to the max-min connectivity recurrence all belong to terms this appendix retires.

Table 4. Historical duration ladder of the retired seven-term configuration. “Cal.” is calibrated macro-scroll Dice; “ $t = .40$ ” uses a fixed threshold; “gain” is calibrated macro gain over the M7 initial model. All calibrated thresholds equal the 0.25 search boundary, so the optimum is censored.

Samples	Cal.	$t = .40$	Gain	Min. gain
1,024	.57944	.57136	+.01968	+.01917
2,048	.57886	.57086	+.01910	+.01868
3,072	.58037	.57253	+.02062	+.02037
4,096	.58022	.57196	+.02047	+.02024
5,120	.57873	.57014	+.01897	+.01813
6,144	.58104	.57329	+.02128	+.02124
7,168	.58293	.57537	+.02317	+.02276
8,192	.58037	.57226	+.02061	+.02031

B A Cautionary Note on Trust Regions

The retired configuration projected the student’s complete parameter vector back into a relative- L^2 ball around the released M7 parameters after every optimizer update:

$$\theta \leftarrow \theta_0 + \min\left(1, \frac{r\|\theta_0\|_2}{\|\theta - \theta_0\|_2}\right)(\theta - \theta_0), \quad r = 0.00275354. \quad (4)$$

We describe it separately from the retired loss terms of Appendix A because its consequences were not local to itself. While it was active, it made the other terms unmeasurable, and every ablation verdict recorded under it is uninformative.

B.1 Why It Seemed Reasonable

The premise was conservative and, stated abstractly, sound. The released M7 is a working expert; the fine teacher is locally more informative but globally incomplete and model-derived; a student

free to move anywhere in parameter space can follow a confident teacher into a region where it is wrong, and the loss will not notice because the teacher supplies the targets. Bounding the student to a neighbourhood of a known-good function is a cheap insurance policy against that failure. The radius was not arbitrary either: it was derived from an earlier measurement of how far the function could safely depart, and inherited across several subsequent versions.

The mistake was not the mechanism. It was retaining a radius across a change of objective, corpus, and optimizer without re-measuring what the new configuration actually required, and then reading the resulting absence of movement as an absence of effect.

B.2 What It Actually Did

The telemetry was unambiguous and was recorded at the time. The projection was active for 97.7% of updates in the first evaluation interval and for 100% of updates in every interval thereafter, with a mean pre-projection step slightly larger than the radius itself. Roughly a quarter of every optimizer step was being discarded. The ball was not a guardrail that occasionally caught an excursion; it was a boundary the run rode continuously, which makes the radius a hyperparameter of the method rather than a safety margin.

The decisive number came later, and from an observation rather than an experiment. Once a wide-corpus run was trained without any projection, we simply measured where it ended up. The recipe that produced the model in this paper sits at roughly 67 times the shipped radius after 30,000 exposures and 82 times after 50,000, and it keeps moving. The displacement the successful method requires is nearly two orders of magnitude outside the ball that earlier versions were confined to. No amount of loss design inside that ball could have found this model, because the model is not in it.

B.3 The Second-Order Damage

A saturated constraint does not merely slow a run down. It silently converts every comparison made under it into a comparison of projections. Two arms differing in one loss term will produce nearly identical functions if the projection removes most of the difference, and the correct reading of “these two runs are indistinguishable” is then “this experiment had no power,” not “this term does nothing.” Our own record contains the clean example: a carefully matched pair testing one mask refinement of the M7 anchor produced probability fields differing by fractions of a percent at every threshold, and the run logs show why—the anchor’s own loss component was mechanically near zero, because the function never left M7’s neighbourhood. That pair was recorded as a null result for the refinement. The refinement is in the shipped objective today, and it is one of the three mask properties in Sec. 4 that the term needs to work.

This is the mechanism by which a saturated trust region manufactures a stack of loss terms. Each new term is added, measured as harmless, and kept; the terms that would have been rejected are protected by the same constraint that prevents the terms that would have been accepted from showing a benefit.

B.4 What We Would Do Instead

We do not conclude that parameter-space trust regions are a bad idea. We conclude that an unmeasured one is, and that the measurement is cheap.

Log the projection’s activity fraction and the pre-projection step norm from the first interval, and treat sustained saturation as a failed run rather than a successful guard. Before trusting any ablation, measure the unconstrained displacement of the same recipe and compare it with the radius; if the free run ends up far outside, the ablation was run at zero power and its verdict should be discarded rather than recorded. Re-derive the radius whenever the objective, corpus, or optimizer changes, since a radius calibrated for one of them carries no authority over another. Where the underlying worry is following a model-derived teacher into its own errors, prefer a constraint expressed in function space and restricted to where the teacher is silent or confidently agrees, which is what Eq. 3 does, and which survived ablation when the parameter-space constraint did not.

C Additive 2D Curve Fitting

This appendix describes the downstream postprocess. It is separated from the body because it is not part of the model: it runs after thresholding, changes no weights, and is measured on its own terms. Keeping the boundary explicit is deliberate. A gap-repair stage that is reported jointly with a segmenter can quietly become a second segmentation method, and an improvement in one stage can then hide a regression in the other.

C.1 Method

The postprocess repairs short omissions in the thresholded predictions. It is additive only: accepted joins set foreground pixels but never erase model output, which makes every intervention localizable and reversible.

For each z-slice, we apply Zhang–Suen thinning [Zhang and Suen 1984], remove short spurs, label the 8-connected skeleton and mask components, and extract curve endpoints with outward tangents. We then snap endpoints to the mask edge so distances describe the visible gap rather than skeleton retraction. Endpoints clipped by the volume or missing data are retained for audit but may not be matched.

We score endpoint pairs using distance, facing, adjacent-plane evidence, and cross-slice track support. Before a pair enters deterministic greedy matching it must satisfy the following requirements:

- (1) it does not close the same skeleton or mask component;
- (2) its tangents face the gap and are sufficiently opposed;
- (3) its reach is at most 21 pixels, while the calibrated safe regime at six pixels or less requires less supporting evidence;
- (4) if an umbilicus is available, its radial displacement is at most three pixels;
- (5) its cubic Bezier has a bounded arc-to-chord ratio and does not approach a third component;
- (6) the path corridor does not cross another sheet on nearby slices; and
- (7) it neither crosses nor touches an already accepted join, and neither endpoint has taken a better match.

A first matching pass seeds endpoint and connection tracks. A second pass follows each new track through z, uses connection confidence, and retains only joins supported across at least three slices. We then paint a cubic Bezier disk stroke and record every added pixel.

C.2 Measured Behaviour

In a calibrated $4 \times 5 \times 5$ study the fitter accepted 481 joins and added 12,697 pixels. Registered high-resolution precision was 92.5% overall and 98.4% for joins no longer than six pixels. We observed no full-turn fusion and measured a 21% reduction in atlas overlap. A larger 21-cube census measured 98.3% weighted precision.

Adjacent high-resolution connectivity is useful confirmation but is not, by itself, proof that a join remains on one wrap. The umbilicus-aware radial gate is our primary cross-wrap safeguard. A release evaluation should therefore report false joins by gate, distance, and radial regime rather than only an aggregate precision.

A bridge scanner reports thin-neck weld candidates, but cutting is disabled by default because erasure breaks the additive contract. An experimental mesh-stage slab splitter separated zero of 246 fused runs; it is retained in the source tree as a measured negative result and is not part of the recommended pipeline.

C.3 Later Postprocessors

Three further opt-in stages are implemented in the repository and evaluated in their own records rather than here: a checked repair profile that recovers more local continuations under a tighter radial guard, a probability-guided continuity tool that reaches beyond the native fitter’s 21-pixel limit, and a stage that extends supported curves into explicit 3D patches. All three share this appendix’s contract—additive, opt-in, separately measured, no model change—and all three currently pin the earlier operating threshold, so each

must be re-qualified before being combined with the shipped inference recipe. None is a production default, and none should be read as a global threshold reduction or an anatomical certification.

References

- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the Knowledge in a Neural Network. In *NIPS Deep Learning and Representation Learning Workshop*. arXiv:1503.02531
- Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, and Klaus H. Maier-Hein. 2021. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods* 18 (2021), 203–211. doi:10.1038/s41592-020-01008-z
- leon (@leyawn). 2015. Socrates and Darryl. Twitter post. <https://twitter.com/leyawn/status/585859882869469184> Posted 2015-04-08.
- Plato. n.d.. Apology, Section 21d. Perseus Digital Library. <https://www.perseus.tufts.edu/hopper/text?doc=Perseus%3Atext%3A1999.01.0170%3Atext%3DApol.%3Asection%3D21d> Accessed 2026-09-01.
- Daniel Rebain, Baptiste Angles, Julien Valentin, Nicholas Vining, Jiju Peethambaran, Shahram Izadi, and Andrea Tagliasacchi. 2019. LSMAT: Least Squares Medial Axis Transform. *Computer Graphics Forum* 38, 6 (2019), 5–18. doi:10.1111/cgf.13599
- Olaf Ronneberger, Philipp Fischer, and Thomas Brox. 2015. U-Net: Convolutional Networks for Biomedical Image Segmentation. In *Medical Image Computing and Computer-Assisted Intervention*. 234–241. doi:10.1007/978-3-319-24574-4_28
- William Brent Seales, Clifford Seth Parker, Michael Segal, Emanuel Tov, Pnina Shor, and Yosef Porath. 2016. From Damage to Discovery via Virtual Unwrapping: Reading the Scroll from En-Gedi. *Science Advances* 2, 9 (2016), e1601247. doi:10.1126/sciadv.1601247
- Suprosanna Shit, Johannes C. Paetzold, Anjany Sekuboyina, Andrey Zhylyka, Ivan Ezhov, Alexander Unger, Jose P. W. Pluim, Ulrich Bauer, and Bjoern H. Menze. 2021. cDice – A Novel Topology-Preserving Loss Function for Tubular Structure Segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 16560–16569.
- Vesuvius Challenge. 2026a. Released M7 Surface Segmentation Model. Model authors, release URL, and version to be finalized before submission.
- Vesuvius Challenge. 2026b. Vesuvius Challenge Data and Tasks. Artifact citation and access date to be finalized before submission.
- Villa Model Authors. 2026. Native-Fine Papyrus Surface Segmentation Teacher. Authoritative software and checkpoint citation to be finalized before submission.
- T. Y. Zhang and C. Y. Suen. 1984. A Fast Parallel Algorithm for Thinning Digital Patterns. *Commun. ACM* 27, 3 (1984), 236–239. doi:10.1145/357994.358023

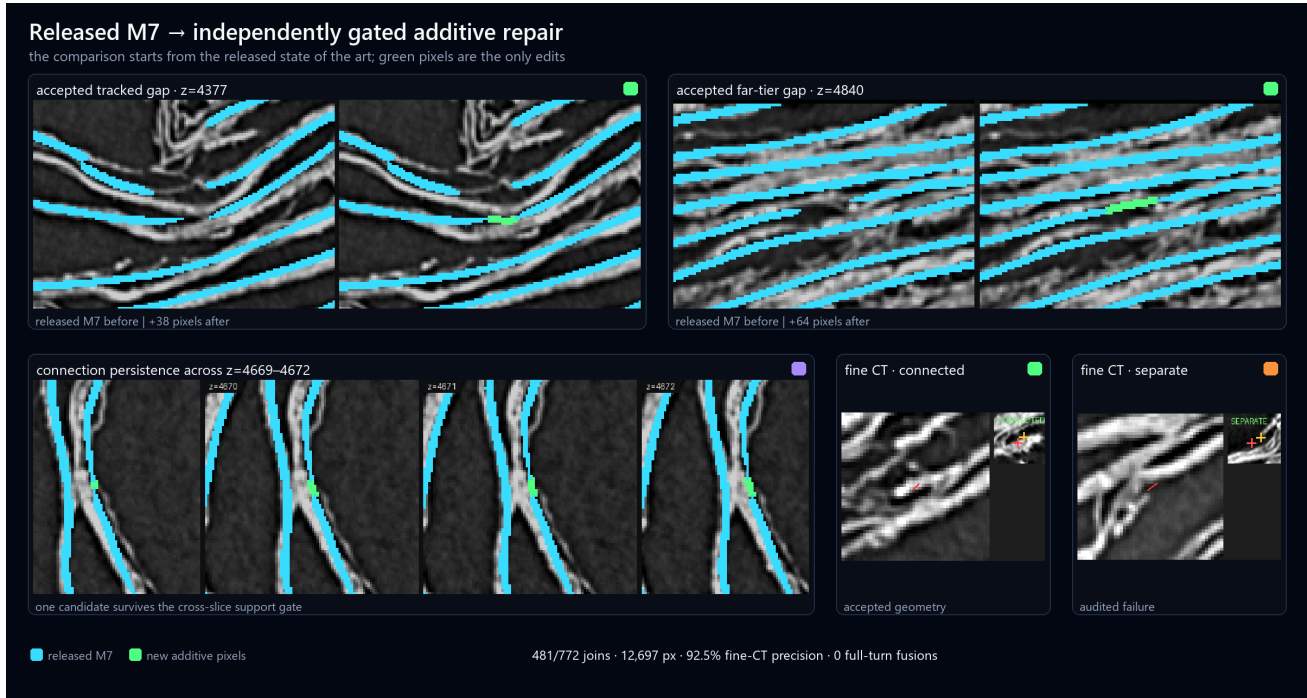


Fig. 8. Measured line-fitter behavior on the released M7 baseline. Left: representative M7 masks before and after additive accepted joins. Center: the same connection persists across neighboring z-slices. Right: registered fine-CT evidence distinguishes a supported connection from a required separation. The calibration accepted 481 of 772 proposals, added 12,697 pixels, reached 92.5% registered fine-CT precision, and produced no observed full-turn fusion. Cyan is released M7 and green is fitted addition.